

Q1 2021 (01 Sep Shift 2)

Let the points of intersections of the lines $x - y + 1 = 0$, $x - 2y + 3 = 0$ and $2x - 5y + 11 = 0$ are the mid points of the sides of a triangle ABC. Then the area of the triangle ABC is .

Q2 2021 (31 Aug Shift 2)

Let A be the set of all points (α, β) such that the area of triangle formed by the points $(5, 6)$, $(3, 2)$ and (α, β) is 12 square units. Then the least possible length of a line segment joining the origin to a point in A, is :

- (1) $\frac{4}{\sqrt{5}}$
- (2) $\frac{16}{\sqrt{5}}$
- (3) $\frac{8}{\sqrt{5}}$
- (4) $\frac{12}{\sqrt{5}}$

Q3 2021 (31 Aug Shift 1)

If p and q are the lengths of the perpendiculars from the origin on the lines,

$$x \operatorname{cosec} \alpha - y \sec \alpha = k \cot 2\alpha \text{ and}$$

$$x \sin \alpha + y \cos \alpha = k \sin 2\alpha$$

respectively, then k^2 is equal to:

- (1) $4p^2 + q^2$
- (2) $2p^2 + q^2$
- (3) $p^2 + 2q^2$
- (4) $p^2 + 4q^2$

Q4 2021 (27 Aug Shift 2)

Let $A(a, 0)$, $B(b, 2b + 1)$ and $C(0, b)$, $b \neq 0$, $b \neq 1$, be points such that the area of triangle ABC is 1sq. unit, then the sum of all possible values of a is :

- (1) $\frac{-2b}{b+1}$
- (2) $\frac{2b}{b+1}$

(3) $\frac{2b^2}{b+1}$

(4) $\frac{-2b^2}{b+1}$

Q5 2021 (27 Aug Shift 1)

Let A be a fixed point $(0, 6)$ and B be a moving point $(2t, 0)$. Let M be the mid-point of AB and the perpendicular bisector of AB meets the y -axis at C . The locus of the mid-point P of MC is :

(1) $3x^2 - 2y - 6 = 0$

(2) $3x^2 + 2y - 6 = 0$

(3) $2x^2 + 3y - 9 = 0$

(4) $2x^2 - 3y + 9 = 0$

Q6 2021 (26 Aug Shift 1)

Let ABC be a triangle with $A(-3, 1)$ and $\angle ACB = \theta$, $0 < \theta < \frac{\pi}{2}$. If the equation of the median through B is $2x + y - 3 = 0$ and the equation of angle bisector of C is $7x - 4y - 1 = 0$, then $\tan \theta$ is equal to:

(1) $\frac{1}{2}$

(2) $\frac{3}{4}$

(3) $\frac{4}{3}$

(4) 2

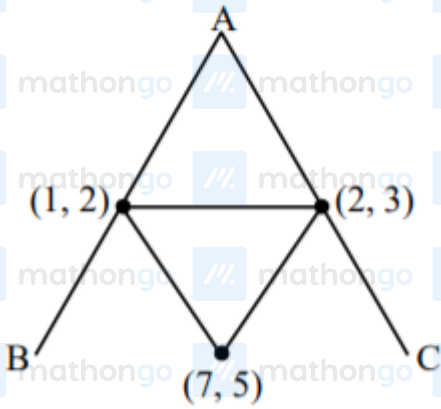
Answer Key

Q1 (6)	Q2 (3)	Q3 (1)	Q4 (4)
Q5 (3)	Q6 (3)		

Q1 (6)

intersection point of give lines are $(1, 2)$, $(7, 5)$,

$(2, 3)$



$$\Delta = \frac{1}{2} \begin{vmatrix} 1 & 2 & 1 \\ 7 & 5 & 1 \\ 2 & 3 & 1 \end{vmatrix}$$

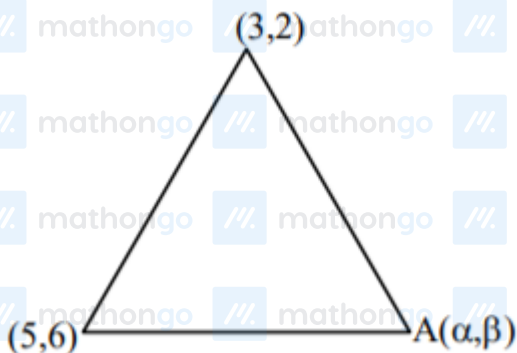
$$= \frac{1}{2} [1(5-3) - 2(7-2) + 1(21-10)]$$

$$= \frac{1}{2} [2 - 10 + 11]$$

$$\Delta_{DEF} = \frac{1}{2} (3) = \frac{3}{2}$$

$$\Delta_{ABC} = 4\Delta_{DEF} = 4 \left(\frac{3}{2} \right) = 6$$

Q2 (3)



$$\begin{vmatrix} 1 & 5 & 6 & 1 \\ 2 & 3 & 2 & 1 \\ \alpha & \beta & 1 & 1 \end{vmatrix} = 12$$

$$4\alpha - 2\beta = \pm 24 + 8$$

$$\Rightarrow 4\alpha - 2\beta = +24 + 8 \Rightarrow 2\alpha - \beta = 16$$

$$2x - y - 16 = 0 \dots (1)$$

$$\Rightarrow 4\alpha - 2\beta = -24 + 8 \Rightarrow 2\alpha - \beta = -8$$

$$2x - y + 8 = 0 \dots (2)$$

perpendicular distance of (1) from (0, 0)

$$\left| \frac{0-0-16}{\sqrt{5}} \right| = \frac{16}{\sqrt{5}}$$

perpendicular distance of (2) from (0, 0) is

$$\left| \frac{0-0+8}{\sqrt{5}} \right| = \frac{8}{\sqrt{5}}$$

Q3 (1)

$$\text{First line is } \frac{x}{\sin \alpha} - \frac{y}{\cos \alpha} = \frac{k \cos 2\alpha}{\sin 2\alpha}$$

$$\Rightarrow x \cos \alpha - y \sin \alpha = \frac{k}{2} \cos 2\alpha$$

$$\Rightarrow p = \left| \frac{k}{2} \cos \alpha \right| \Rightarrow 2p = |k \cos 2\alpha| \dots (i)$$

$$\text{second line is } x \sin \alpha + y \cos \alpha = k \sin 2\alpha$$

$$\Rightarrow q = |k \sin 2\alpha| \dots (ii)$$

$$\text{Hence } 4p^2 + q^2 = k^2 \quad (\text{From (i) \& (ii)})$$

Q4 (4)

$$\begin{vmatrix} 1 & a & 0 & 1 \\ 2 & b & 2b+1 & 1 \\ 0 & b & b & 1 \end{vmatrix} = 1$$

$$\Rightarrow \begin{vmatrix} a & 0 & 1 \\ b & 2b+1 & 1 \\ 0 & b & 1 \end{vmatrix} = \pm 2$$

$$\Rightarrow a(2b+1-b) - 0 + 1(b^2 - 0) = \pm 2$$

$$\Rightarrow a = \frac{\pm 2 - b^2}{b+1}$$

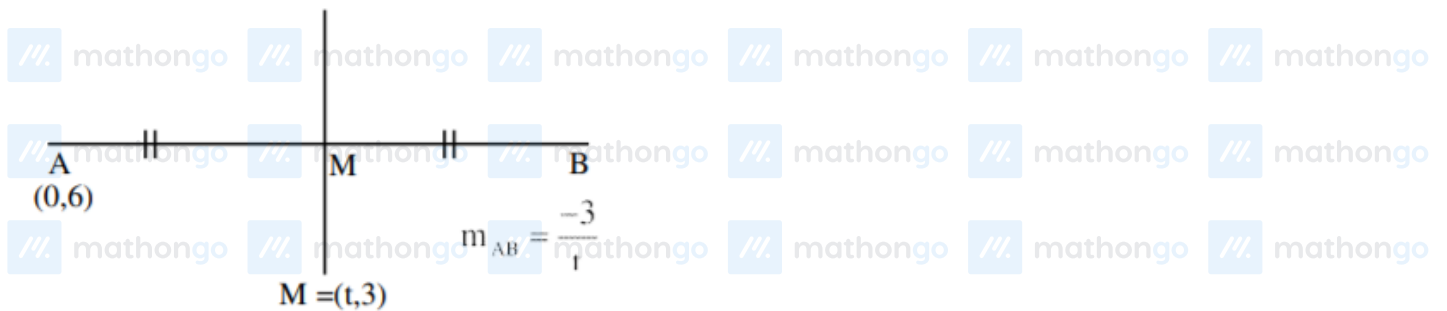
$$\therefore a = \frac{2-b^2}{b+1} \text{ and } a = \frac{-2-b^2}{b+1}$$

sum of possible values of 'a' is

$$= \frac{-2b^2}{a+1} \text{ Ans.}$$

Q5 (3)

A(0, 6) and B(2t, 0)



Perpendicular bisector of AB is

$$(y - 3) = \frac{t}{3}(x - t)$$

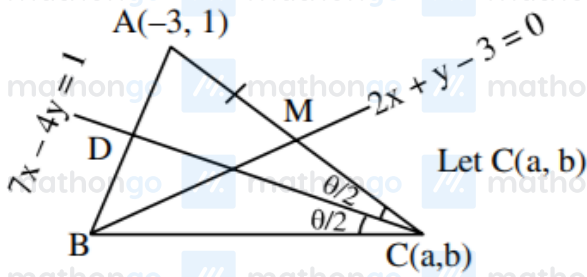
$$\text{So, } C = \left(0, 3 - \frac{t^2}{3}\right)$$

Let P be (h, k)

$$h = \frac{t}{2}; k = \left(3 - \frac{t^2}{6}\right)$$

$$\Rightarrow k = 3 - \frac{4h^2}{6} \Rightarrow 2x^2 + 3y - 9 = 0 \text{ option (3)}$$

Q6 (3)



$$\therefore M\left(\frac{a-3}{2}, \frac{b+1}{2}\right) \text{ lies on } 2x + y - 3 = 0$$

$$\Rightarrow 2a + b = 11 \dots\dots\dots(i)$$

$$\therefore C \text{ lies on } 7x - 4y = 1$$

$$\Rightarrow 7a - 4b = 1 \dots\dots\dots(ii)$$

$$\therefore \text{by (i) and (ii) : } a = 3, b = 5$$

$$\Rightarrow C(3, 5)$$

$$\therefore m_{AC} = 2/3$$

$$\text{Also, } m_{CD} = 7/4$$

$$\Rightarrow \tan \frac{\theta}{2} = \left| \frac{\frac{2}{3} - \frac{7}{4}}{1 + \frac{14}{12}} \right| \Rightarrow \tan \frac{\theta}{2} = \frac{1}{2}$$

$$\Rightarrow \tan \theta = \frac{2 \cdot \frac{1}{2}}{1 - \frac{1}{4}} = \frac{4}{3}$$